Expt No. 2 **Dt:**

DETERMINATION OF WORK DONE FOR DIFFERENT THERMODYNAMIC PROCESSES

Aim: To write and execute python program for determination of work done for different thermodynamic processes.

Interactive Development Environment (IDE) Used : Jupyter NoteBook

Programming Language Used : Python

Basic theory :.

1. Isobaric Process (Constant Pressure)

- Pressure remains constant throughout the process. (P = Constant)
- Work Done (W)

  $W = P (V2 - V1)$

  Where

  P = Pressure (kPa)

  V1 = Initial Volume (m^3)

  V2 = Final Volume (m^3)

  W = Work Done (kJ)

2. Isochoric Process (Constant Volume)

- Volume remains constant throughout the process. (V = Constant, V2 = V1)
- Work Done (W)

  $W = 0$

  Where

  V1 = Initial Volume (m^3)

  V2 = Final Volume (m^3)

  W = Work Done (kJ)

3. Isothermal Process (Constant Temperature)

- Temperature remains constant throughout the process. (P1V1 = P2V2)
- Final Volume (V2)

  $V2 = (P1 \times V1) / P2$
- Work Done (W)

  $W = P1V1 \ln(V2/V1)$

  Where

  P1 = Initial Pressure (kPa)

  P2 = Final Pressure (kPa)

  V1 = Initial Volume (m^3)

  V2 = Final Volume (m^3)

  W = Work Done (kJ)

4. Polytropic Process

- PV^n = Constant ($n \neq 0$ and $n \neq 1$)

- Final Volume (V2)
  $V2 = V1\,(P1/P2)^{(1/n)}$
- Work Done (W)
  $W = (P2\,V2 - P1\,V1)/(1 - n)$
  Where
  P1 = Initial Pressure (kPa)
  P2 = Final Pressure (kPa)
  V1 = Initial Volume (m^3)
  V2 = Final Volume (m^3)
  n = Polytropic Exponent
  W = Work Done (kJ)
- Special Cases
  n = 0 : Isobaric Process
  n = 1 : Isothermal Process

Python Concepts used:

Python Command	**Description**	**General Example**
import	Imports a module into the program.	import math
print()	Displays text or values on the screen.	print("Welcome")
input()	Accepts input from the user.	name = input("Enter your name: ")
int()	Converts the input into an integer.	age = int(input("Enter Age: "))
float()	Converts the input into a floating-point number.	salary = float(input("Enter Salary: "))

if...else	Executes one block if the condition is true; otherwise executes another block.	if marks >= 35:
		    print("Pass")
		else:
		    print("Fail")

if...elif...else	Selects one block from multiple alternatives based on conditions.	if n > 0:
		    print("Positive")
		elif n < 0:
		    print("Negative")
		else:
		    print("Zero")

for loop	Repeats a block of code for a specified number of iterations.	for i in range(5): print(i)
range()	Generates a sequence of integers for iteration in a loop.	**Stop value only:** range(5) → 0, 1, 2, 3, 4 **Start and Stop values:** range(2, 7) → 2, 3, 4, 5, 6 **Start, Stop and Step values:** range(2, 11, 2) → 2, 4, 6, 8, 10

continue	Skips the remaining statements in the current iteration and proceeds to the next iteration.	for i in range(5):
		    if i == 2:
		        continue
		        print(i)
		
		Output: 0 1 3 4

math.log()	Computes the natural logarithm (ln) of a number.	math.log(10)
round()	Rounds a number to the specified number of decimal places.	round(3.14159, 2) → 3.14

Python Program:

```
# Program to Calculate Work Done in Thermodynamic Processes
import math
print("THERMODYNAMIC PROCESSES")
print("-----------------------")
print("1. Isobaric Process")
print("2. Isochoric Process")
print("3. Isothermal Process")
print("4. Polytropic Process")
choice = int(input("\nEnter your choice (1-4): "))
if choice == 1:
      P = float(input("Enter Pressure (kPa): "))
      V1 = float(input("Enter Initial Volume (m^3): "))
      V2 = float(input("Enter Final Volume (m^3): "))
      W = P * (V2 - V1)
      print("Work Done =", round(W,2), "kJ")
elif choice == 2:
      print("Volume remains constant.")
      print("Work Done = 0 kJ")
elif choice == 3:
      P1 = float(input("Enter Initial Pressure (kPa): "))
      P2 = float(input("Enter Final Pressure (kPa): "))
      V1 = float(input("Enter Initial Volume (m^3): "))
      V2 = (P1 * V1) / P2
      W = P1 * V1 * math.log(V2 / V1)
      print("Final Volume =", round(V2,3), "m^3")
      print("Work Done =", round(W,2), "kJ")
elif choice == 4:
      P1 = float(input("Enter Initial Pressure (kPa): "))
      P2 = float(input("Enter Final Pressure (kPa): "))
      V1 = float(input("Enter Initial Volume (m^3): "))
      print("n\tFinal Volume\tWork Done")
      for i in range(0,6):
            n = i * 0.5
            if n == 0:
                  print(n, "Isobaric Case")
                  continue
            elif n == 1:
                  print(n, "Isothermal Case")
                  continue
            V2 = V1 * (P1 / P2) ** (1 / n)
            W = (P2 * V2 - P1 * V1) / (1 - n)
            print(round(n,1), "\t", round(V2,3), "\t", round(W,2))
else:
      print("Invalid Choice")
```

Sample Input and Output :

Result :

Python program was written and executed to determine the work done for different thermodynamic processes.